INCH-POUND

MIL-DTL-17/74C
w/AMENDMENT 1
13 April 2010

SUPERSEDING
MIL-C-17/74C
18 July 1985

DETAIL SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, FLEXIBLE, COAXIAL, 50 OHMS, UNARMORED M17/74-RG213, AND ARMORED M17/74-RG215

Inactive for new design after 13 August 1993. For new design use MIL-C-17/189.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17.

NOTE: This cable uses PVC material and is not to be used in enclosed environments or shipboard applications. The replacements that are to be used in enclosed areas or shipboard applications are referenced in the following table.

The Air Force has restricted use of PVC in aerospace and ground support applications.

Cables listed on the current QPL may continue to be manufactured and supplied for existing enclosed applications only for a period not to exceed 3 years from the date of this specification.

TABLE I. Cross-reference data.

Current Part or Identifying Number (PIN)	Replacement PIN
M17/74-RG213	M17/189-00001
M17/74-RG215	M17/189-00002

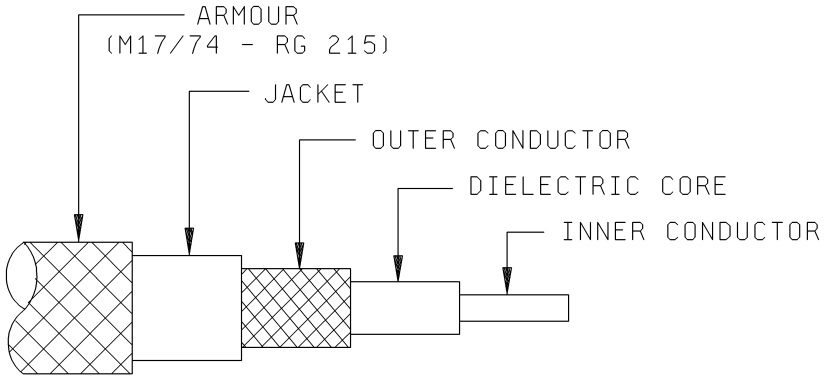


FIGURE 1. Configuration.

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TABLE II. Description.

Components	Construction details
Inner conductor	Seven strands of bare copper wire, each strand 0.0296 inch diameter. Overall diameter: 0.0888 inch ± 0.0010.
Dielectric core	Type A-1: Solid, polyethylene. Diameter: 0.285 inch ± 0.007.
Outer conductor	Double braid of AWG #33, bare copper wire. Diameter: 0.340 inch maximum. Coverage: 95.3% nominal Carriers: 24 Ends: 8 Picks/inch: 6.5 ± 10%
Jacket	Type IIa: PVC. Diameter: 0.405 inch ± 0.007.
Armored M17/74-RG215	Single braid of aluminum-alloy wire. Diameter: 0.475 inch maximum.

ENGINEERING INFORMATION

Continuous working voltage: 3,700 V rms, maximum.

Operating frequency: 1 GHz, maximum.

Velocity of propagation: 65.9 percent, nominal.

Power rating: See figure 2.

Operating temperature range: -40°C to +85°C.

Inner conductor properties:

DC resistance (maximum at 20°C): 0.176 ohm per 100 feet.

Elongation: 25 percent, minimum.

Tensile strength: Not applicable.

Engineering notes: This cable useful in general purpose medium low temperature applications. (See connector series “N”, “C” and “SC” in accordance with MIL-PRF-39012. For M17/74-RG213, NATO preferred type NWR-1; for M17/74-RG215, NATO preferred type NWR-17.)

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REQUIREMENTS

Dimensions, configuration, and description: See figure 1 and table II.

Environmental and mechanical:

Visual and mechanical examination:

Out-of-roundness: Not applicable.

Eccentricity: 10 percent, maximum.

Adhesion of conductors:

Inner conductor to core: 9 pounds, minimum; 90 pounds, maximum.

Aging stability: +98ºC ± 2ºC.

Stress crack resistance: Not applicable.

Outer conductor integrity: Not applicable.

Cold bend: -40ºC ± 2ºC.

Dimensional stability: +85ºC ±2ºC.

Inner from conductor core: 0.062 inch, maximum.

Inner conductor from jacket: 0.125 inch, maximum.

Contamination: Applicable.

Bendability: Not applicable.

Flammability: Not applicable.

Weight:

M17/74-RG213: 0.111 pound per foot, maximum.
M17/74-RG215: 0.138 pound per foot, maximum.

Electrical:

Continuity: Applicable.

Spark test: 5,000 V rms, +10%, -0%.

Voltage withstanding: 10,000 V rms, +10%, -0%.

Insulation resistance: Not applicable.

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Corona extinction voltage: 5,000 V rms, minimum.

Characteristic impedance: 50 ohms ± 2.

Attenuation: See figure 2.

Structural return loss: See figure 3.

Capacitance: 32.2 pF per foot, maximum.

Capacitance stability: Not applicable.

Capacitance unbalance: Not applicable.

Transmission unbalance: Not applicable.

Mechanically induced noise voltage: Not applicable.

Time delay: Not applicable.

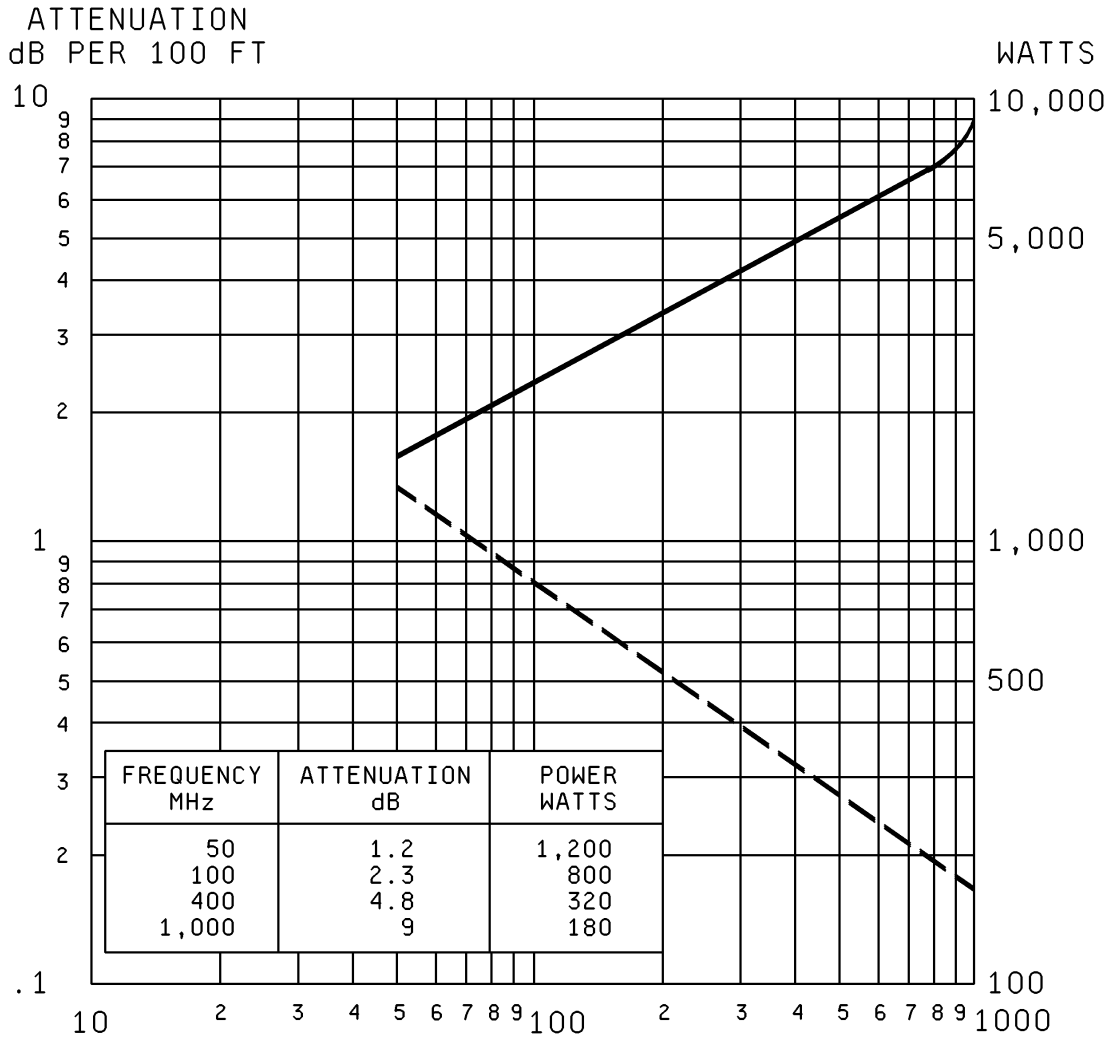
PIN: M17/74-RG213 and M17/74-RG215. See table I.

Supersession date: See table III.

TABLE III. Cross-reference of PIN.

PIN	Superseded PIN
M17/74/RG213	----------
M17/74-RG215	RG-215/U; RF-10A/U per MIL-C-17/5 (canceled)

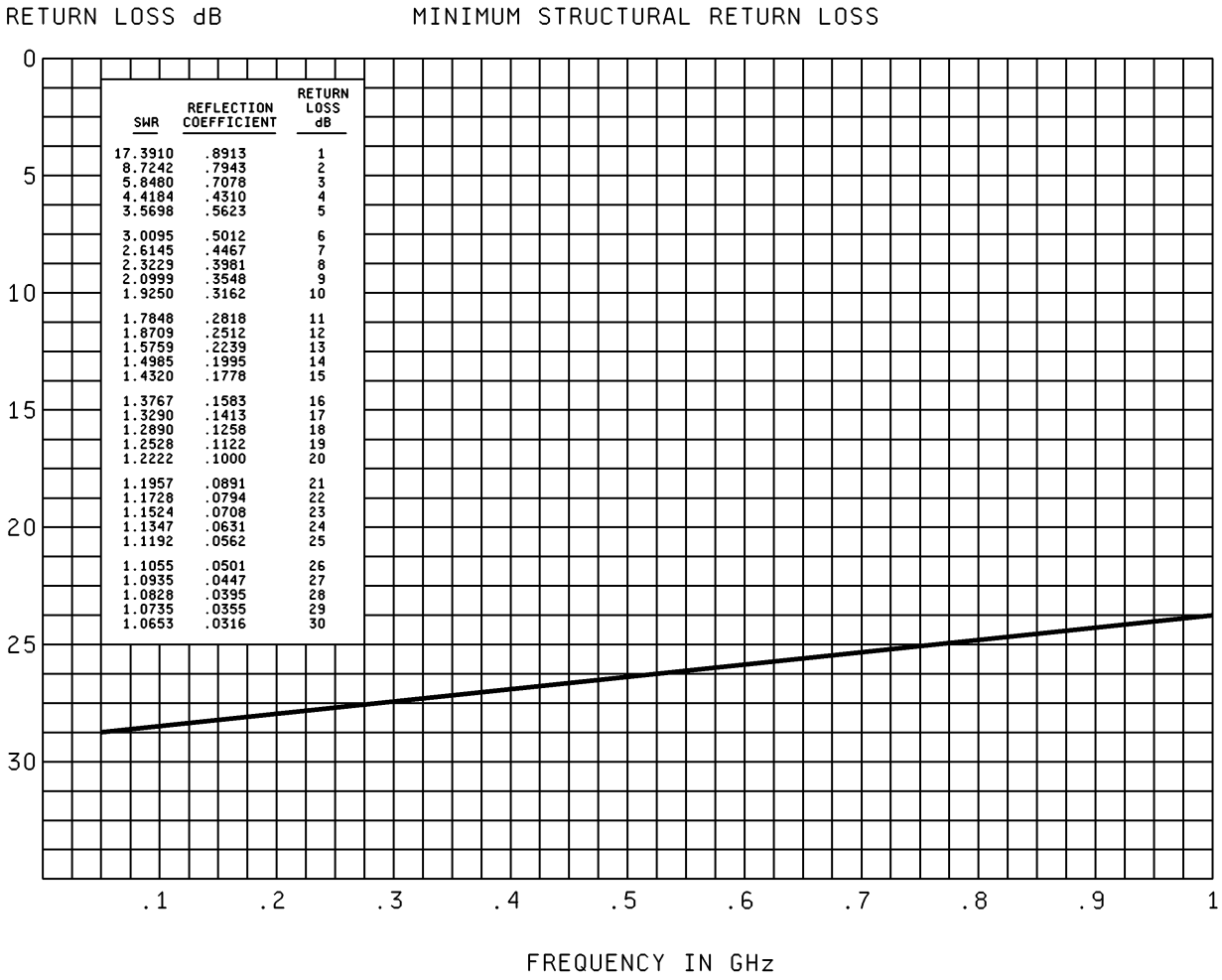
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FREQUENCY MHz	ATTENUATION dB	POWER WATTS
50	1.2	1,200
100	2.3	800
400	4.8	320
1,000	9	180

FIGURE 2. Power rating and attenuation.

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SWR	REFLECTION COEFFICIENT	RETURN LOSS dB
17.3910	.8913	1
8.7242	.7943	2
5.8480	.7078	3
4.4184	.4310	4
3.5698	.5623	5
3.0095	.5012	6
2.6145	.4467	7
2.3229	.3981	8
2.0999	.3548	9
1.9250	.3162	10
1.7848	.2818	11
1.8709	.2512	12
1.5759	.2239	13
1.4985	.1995	14
1.4320	.1778	15
1.3767	.1583	16
1.3290	.1413	17
1.2890	.1258	18
1.2528	.1122	19
1.2222	.1000	20
1.1957	.0891	21
1.1728	.0794	22
1.1524	.0708	23
1.1347	.0631	24
1.1192	.0562	25
1.1055	.0501	26
1.0935	.0447	27
1.0828	.0395	28
1.0735	.0355	29
1.0653	.0316	30

FIGURE 3. Structural return loss.

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Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-17, this document references the following:

MIL-PRF-39012
MIL-C-17/189

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 6145-2010-028)

Review activities:
Army – AT, CR4, MI
Navy – AS, MC, OS, SH, TD
Air Force – 19, 99
DLA - IS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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